

BIZEV-02: Instrumentation

Series: BIZEV — Business Events & Funnel Analysis | **Notebook:** 2 of 7 | **Created:** March 2026 | **Last Updated:** 07/15/2026

Overview

Capturing meaningful business events requires deliberate instrumentation. This notebook covers the practical techniques for generating business events — from OneAgent auto-detection of web request data, to custom API ingestion, SDK integration, and OpenTelemetry span-to-bizevent mapping. You will also learn best practices for event naming conventions, payload design, and cardinality management to keep your business analytics clean and performant.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
Permissions	<code>storage:bizevents:read</code> , <code>bizevents.ingest</code>
OneAgent	Deployed on at least one application host (for auto-detection)
Knowledge	BIZEV-01 fundamentals, basic REST API concepts

No Grail yet? Every capture path in this notebook writes to the Grail `bizevents` table and therefore requires a SaaS tenant with Grail. For what a classic (Gen2) environment can do instead — request attributes, session/action properties + USQL, log metrics — and for the hybrid pattern that adopts these capture paths with only a minimal Gen3 footprint, see **BIZEV-07: Gen2 vs Gen3 — Business Events Without (or Before) the Full Move**.

1. OneAgent Auto-Detection

OneAgent can automatically capture business events from web application requests using **capture rules**. This is the lowest-effort approach — no code changes required.

Setting Up Capture Rules

Navigate to **Settings > Business Analytics > OneAgent > Capture rules** and define:

Setting	Description	Example
Trigger	URL pattern or request attribute match	<code>/api/v1/orders/*</code>
Event type	The <code>event.type</code> value to assign	<code>com.myapp.order.created</code>
Data extraction	Request/response attributes to capture	<code>order_id</code> , <code>total_amount</code>
Filtering	Conditions to limit capture	<code>HTTP method == POST</code>

What Gets Captured Automatically

When a capture rule matches, OneAgent creates a business event with:

- The configured `event.type`
- `event.provider` set to the application name
- Extracted request attributes as custom payload fields
- Automatic correlation with the trace context (span ID, trace ID)

Tip: Start with auto-detection for existing web applications, then add custom instrumentation for backend processes that don't have HTTP endpoints.

```
// Check which business events are being captured by OneAgent auto-detection
// Events from OneAgent typically have the application name as event.provider
fetch bizevents, from:-24h
| filter isNotNull(dt.entity.service)
| summarize event_count = count(), by:{event.type, event.provider}
| sort event_count desc
| limit 15
```

2. Business Events API Ingestion

The Business Events API allows you to send events from any system — backend services, batch processes, third-party integrations, or IoT devices.

CloudEvents Format

The API expects events in [CloudEvents](#) format:

```
{
  "specversion": "1.0",
  "id": "evt-20260312-001",
  "type": "com.myapp.payment.processed",
  "source": "payment-service",
  "time": "2026-03-12T10:30:00Z",
  "data": {
    "payment_id": "PAY-98765",
    "amount": 249.99,
    "currency": "USD",
    "method": "credit_card",
    "customer_tier": "gold"
  }
}
```

API Endpoint

```
POST https://{environment-id}.live.dynatrace.com/api/v2/bizevents/ingest
Content-Type: application/cloudevents+json
Authorization: Api-Token dt0c01.xxxxx
```

Batch Ingestion

For high-volume scenarios, send multiple events in a single request using `application/cloudevents-batch+json` :

```
[
  {"specversion": "1.0", "type": "com.myapp.event1", "source": "svc", "data": {}},
  {"specversion": "1.0", "type": "com.myapp.event2", "source": "svc", "data": {}}
]
```

Important: The API has rate limits. For sustained high throughput (>1000 events/second), use batch ingestion and implement retry logic with exponential backoff.

```
// Verify API-ingested events are arriving
// API-ingested events often have a custom source/provider value
fetch bizevents, from:-1h
| summarize {event_count = count(),
               latest = max(timestamp),
               earliest = min(timestamp)}, by:{event.provider}
| sort event_count desc
```

3. SDK Integration

The Dynatrace OneAgent SDK provides language-specific methods to send business events directly from application code.

Java Example

```
import com.dynatrace.oneagent.sdk.api.BusinessEventsOneAgentApi;

Map<String, Object> attributes = new HashMap<>();
attributes.put("order_id", "ORD-12345");
attributes.put("total", 149.99);
attributes.put("items", 3);

BusinessEventsOneAgentApi.sendBizEvent(
    "com.myapp.checkout.completed",
    attributes
);
```

JavaScript / Node.js Example

```
const dtrum = window.dtrum;

dtrum.sendBizEvent('com.myapp.add-to-cart', {
  product_id: 'PROD-456',
  product_name: 'Widget Pro',
  price: 29.99,
  quantity: 2
});
```

Python Example

```
import oneagent

sdk = oneagent.get_sdk()
sdk.send_biz_event(
    event_type="com.myapp.signup.completed",
    attributes={
        "user_id": "USR-789",
        "plan": "premium",
        "source_campaign": "spring-2026"
    }
)
```

Tip: SDK-generated events are automatically correlated with the active PurePath, linking business events to the full distributed trace.

4. OpenTelemetry Span-to-Bizevent Mapping

If your application already produces OpenTelemetry spans with business-relevant attributes, you can map them to business events using **OpenPipeline processing rules**.

How It Works

1. Application emits OTel spans with business attributes (e.g., `order.id`, `order.total`)
2. OpenPipeline receives the span data
3. A processing rule extracts business attributes and routes them to `bizevents`
4. The original span remains in `spans` — the business event is an additional record

OpenPipeline Configuration

```
# Example OpenPipeline rule for span-to-bizevent mapping
pipelines:
  - name: "Extract checkout events from spans"
    processing:
      - type: bizevent
        condition: "span.name == 'checkout.complete'"
        eventType: "com.myapp.checkout.completed"
        attributes:
          - source: "order.id"
            target: "order_id"
          - source: "order.total"
            target: "amount"
```

Note: This approach avoids double-instrumentation. If your spans already carry business data, use OpenPipeline rather than adding SDK calls.

```
// Look for business events that may have originated from spans
// These often have trace/span correlation fields
fetch bizevents, from:-24h
| filter isNotNull(trace_id) or isNotNull(span_id)
| summarize correlated_count = count(), by:{event.type}
| sort correlated_count desc
| limit 10
```

5. Event Naming Best Practices

A consistent naming convention is critical for long-term analytics. Poor naming leads to fragmented data and unreliable queries.

Recommended Naming Convention

Use reverse-domain notation with a verb suffix:

```
com.<company>.<domain>.<action>;
```

Pattern	Example	Description
<code>com.myapp.order.created</code>	Order placed	New order in system
<code>com.myapp.order.completed</code>	Order fulfilled	Order shipped/delivered
<code>com.myapp.payment.processed</code>	Payment received	Successful payment
<code>com.myapp.payment.failed</code>	Payment rejected	Failed payment attempt
<code>com.myapp.user.signup</code>	New registration	User account created
<code>com.myapp.user.login</code>	Authentication	Successful login
<code>com.myapp.cart.updated</code>	Cart change	Item added/removed

Naming Anti-Patterns

Avoid	Why	Better
<code>order</code>	Too vague — created? cancelled?	<code>com.myapp.order.created</code>
<code>OrderCreated</code>	Inconsistent casing	<code>com.myapp.order.created</code>
<code>com.myapp.order.created.v2</code>	Version in name fragments data	Use payload versioning
<code>event_12345</code>	Meaningless identifier	Use descriptive domain names

```
// Audit event naming – find event types that may need standardization
fetch bizevents, from:-7d
| summarize {event_count = count(),
              first_seen = min(timestamp),
              last_seen = max(timestamp)}, by:{event.type}
| sort event_count desc
```

6. Payload Design and Cardinality

The custom attributes you include in business events determine what analytics are possible. Design payloads carefully.

Payload Design Principles

Principle	Guidance
Include business identifiers	<code>order_id</code> , <code>customer_id</code> , <code>session_id</code>
Include measurable values	<code>amount</code> , <code>quantity</code> , <code>duration_ms</code>
Include categorical dimensions	<code>category</code> , <code>region</code> , <code>customer_tier</code>
Avoid high-cardinality strings	Don't include full URLs, stack traces, or free text
Use consistent types	Always send <code>amount</code> as a number, not sometimes as a string

Cardinality Management

High-cardinality fields (fields with millions of unique values) can degrade query performance.

Field Type	Cardinality	Impact
<code>customer_tier</code> (gold/silver/bronze)	Low (~3 values)	Excellent for <code>summarize by:</code>
<code>product_category</code>	Medium (~100 values)	Good for grouping
<code>order_id</code>	High (millions)	Fine for lookup, avoid <code>summarize by:</code>
<code>full_url</code>	Very high	Avoid — parse into structured fields instead

Warning: Using `summarize by:{order_id}` on millions of unique order IDs will produce a massive result set and slow queries. Use high-cardinality fields for filtering (`filter order_id == "ORD-123"`) rather than grouping.

```
// Assess cardinality of common fields in your business events
fetch bizevents, from:-24h
| summarize {total = count(),
              distinct_types = countDistinct(event.type),
              distinct_providers = countDistinct(event.provider),
              distinct_categories = countDistinct(event.category)}
```

7. Verifying Instrumentation with DQL

After setting up instrumentation, validate that events are arriving correctly and contain the expected data.

```
// Check for gaps in business event ingestion over the last 24 hours
// A healthy pipeline should show consistent event flow
fetch bizevents, from:-24h
| makeTimeseries event_count = count(), interval:15m
```

```
// Validate that required fields are populated (not null)
fetch bizevents, from:-1h
| summarize {total = count(),
              has_type = countIf(isNotNull(event.type)),
              has_provider = countIf(isNotNull(event.provider)),
              has_category = countIf(isNotNull(event.category))}
| fieldsAdd type_pct = round(toDouble(has_type) / toDouble(total) * 100,
                             decimals: 1),
              provider_pct = round(toDouble(has_provider) / toDouble(total) *
                                   100, decimals: 1),
              category_pct = round(toDouble(has_category) / toDouble(total) *
                                   100, decimals: 1)
```

8. Summary and Next Steps

In this notebook, you learned:

- **OneAgent auto-detection** — Capture business events from web requests without code changes
- **API ingestion** — Send events from any system using CloudEvents format
- **SDK integration** — Emit events directly from application code with trace correlation
- **OpenTelemetry mapping** — Route span data to bizevents via OpenPipeline

- **Naming conventions** — Use reverse-domain notation with action verbs
- **Payload design** — Balance richness with cardinality management

Next Steps

- **BIZEV-03: Funnel Analysis** — Use your instrumented events to build conversion funnels
- **BIZEV-05: KPIs and Metrics** — Extract business KPIs from event data

References

- [Business Events API](#)
- [OneAgent SDK for Business Events](#)
- [OpenPipeline Processing](#)

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